

Duration – 70 min.

Q.1 Fill in the blanks:

- (i) The process of automatic promotion is known as widening.
- (ii) In java finalize() method called by garbage collection.
- (iii) A boolean literal represents a logical value true and false in java.
- (iv) Byte code is executed by JVM.

Q.2 Consider the following program and what will be the outcome of each. Also justify your answer. 6

```
(i) public class Test1 {
    public static void main
    (String [] args) {
        int a=2,i=1,j;
        if(a<1 & ++a<=3)
            System.out.println(a);
        System.out.println(a);
        System.out.println(-6);
        j = i++ + ++i + ++i + i++ +
            i--;
        System.out.println("i=" + i);
        System.out.println("j=" + j);
    }
}
```

j = 15

```
(ii) public class Test
    static int i=20;
    public static void main
    (String [] args) {
        int i;
        Test t = new Test();
        System.out.println("i=" + i);
        System.out.println("i="
            + Test.i);
        System.out.println("i=" +
            t.i);
    }
}
```

```
(iii) class Test {public static void main(String
    []ar)
    { Object o=new Object();
    System.out.println( o instanceof String); // false
    int x=5;
    do
    { x++;
    System.out.println(x);
    if(++x<10)
    continue;
    x--;
    System.out.println(x);
    } while(++x<15); } }
```

child class

6
9
11
13
15

Q.3 Write a program to check whether the entered number is buzz or not.

Q.4 Write a Java program to count armstrong numbers between a given interval.

5

5

Start or end
do not